

STATIC INVENTORY PROBLEMS UNDER RISK

UNIVERSITY NUMERICALS WITH EXPLICIT COST-TABLE CONSTRUCTION

PROBLEM 1: DISCRETE DEMAND – EXPECTED COST METHOD

A newspaper vendor must decide how many copies to order. Possible demand and probabilities are:

Demand (D): 40, 50, 60
Probability: 0.2, 0.5, 0.3

Purchase cost = ₹5 per copy
Selling price = ₹8 per copy
Unsold copies have no value.

STEP 1: IDENTIFY COSTS

Shortage cost (C_u) = Selling price – Purchase cost = $8 - 5 = ₹3$ per copy
Excess cost (C_o) = Purchase cost = ₹5 per copy

STEP 2: DECISION ALTERNATIVES

Possible order quantities $Q = 40, 50, 60$

STEP 3: COST RULES

If $Q > D$, Cost = $(Q - D) \times C_o$
If $Q < D$, Cost = $(D - Q) \times C_u$
If $Q = D$, Cost = 0

STEP 4: CONSTRUCT TOTAL COST TABLE

Order Q \ Demand D	40	50	60
40	0	$(50-40) \times 3 = 30$	$(60-40) \times 3 = 60$
50	$(50-40) \times 5 = 50$	0	$(60-50) \times 3 = 30$
60	$(60-40) \times 5 = 100$	$(60-50) \times 5 = 50$	0

STEP 5: EXPECTED COST CALCULATION

For $Q = 40$:
 $E(C) = (0.2 \times 0) + (0.5 \times 30) + (0.3 \times 60) = ₹33$

For $Q = 50$:
 $E(C) = (0.2 \times 50) + (0.5 \times 0) + (0.3 \times 30) = ₹19$

For $Q = 60$:
 $E(C) = (0.2 \times 100) + (0.5 \times 50) + (0.3 \times 0) = ₹45$

STEP 6: OPTIMAL DECISION

Minimum expected cost = ₹19
Optimal order quantity = 50 copies

PROBLEM 2: TOTAL COST MATRIX AND OPPORTUNITY COST

A shopkeeper considers ordering 50, 100, or 150 units. Possible demand is 50, 100, or 150 units.

Shortage cost = ■6 per unit

Excess cost = ■2 per unit

STEP 1: COST RULES

If $Q > D$, Cost = $(Q - D) \times 2$

If $Q < D$, Cost = $(D - Q) \times 6$

STEP 2: TOTAL COST MATRIX

Order Q \ Demand D | 50 | 100 | 150

50 | 0 | $(100-50) \times 6 = 300$ | $(150-50) \times 6 = 600$

100 | $(100-50) \times 2 = 100$ | 0 | $(150-100) \times 6 = 300$

150 | $(150-50) \times 2 = 200$ | $(150-100) \times 2 = 100$ | 0

STEP 3: OPPORTUNITY COST MATRIX

Minimum cost in each column = 0

Opportunity cost = Actual cost – Minimum cost

Order Q \ Demand D | 50 | 100 | 150

50 | 0 | 300 | 600

100 | 100 | 0 | 300

150 | 200 | 100 | 0

STEP 4: MINIMAX REGRET DECISION

Maximum regret for $Q=50 \rightarrow 600$

Maximum regret for $Q=100 \rightarrow 300$

Maximum regret for $Q=150 \rightarrow 200$

Optimal decision = Order 150 units